JESSICA ANDREW

BSc Computer Science (3rd Year) | Graduating November 2026
+27 62 766 3833 • jess.andrew2005@gmail.com • Pretoria, Gauteng • LinkedIn • GitLab

PROFESSIONAL SUMMARY

Dedicated BSc Computer Science student with a strong technical foundation in software development, data structures, algorithms, and problem-solving. Proficient in Java, C++, Python, C# and JavaScript, with hands-on experience in full-stack development (React, Flask, PostgreSQL, Flutter), UI test automation (Playwright, NUnit) and containerised deployment (Docker). Passionate about cloud computing, cybersecurity and software architecture. Strong communicator with proven ability to collaborate effectively with diverse stakeholders.

EDUCATION

Bachelor of Science in Computer Science | Akademia University, Pretoria *Expected: November 2026*
Currently in 3rd and final year of degree program

Completed Modules:

- Advanced Algorithms & Data Structures Optimization
- Software Engineering & Design Patterns
- Project Management & Software Development Lifecycle
- Systems Analysis & Design – feasibility analysis, cost estimation, business cases and agile planning
- Advanced Database Design & SQL Query Optimization
- Full-Stack Web Development (Frontend & Backend Architecture)
- Computer Networks – packet switching, protocol stack, transport layer (TCP/UDP), network layer control plane, link layer & LANs, wireless and mobile networks
- Cybersecurity – CIA triad, threat actors, security strategy and practical security awareness; group analysis of a real-world cyber incident
- Cloud Computing Architecture (AWS, Azure, Google Cloud)
- Mobile Application Development (iOS/Android)
- DevOps & Continuous Integration/Deployment
- Systems Design & Scalability

Recent Semester – New Modules:

- Operating Systems – In-depth study of OS fundamentals including CPU scheduling, memory management, process synchronisation, and processor architecture
- Software Architecture – Software and API design, microservices and API gateways, message queues (RabbitMQ), Docker Compose, architectural diagramming, and system-level design documentation

Capstone Team Project – Symposium Management System (AKBs), team UnityStack:

- Full-stack academic conference management platform with a React web app, Flutter mobile app and Flask REST API (PostgreSQL, JWT, WebSockets), deployed with Docker Compose and Caddy on a VPS
- Developed the reviewer workflow module: multi-role pipeline (Student, Reviewer, Admin) with token-based email links, institution conflict checks and scoring criteria
- Built the AI chatbot in the mobile app: database pattern matching with a Google Gemini API fallback, using multi-model retry with exponential backoff
- Contributed to the React frontend and the GitLab workflow (merge requests, conflict resolution, release tagging)

National Senior Certificate (Matric)
Completed 2023 | Key Subjects: Mathematics, Engineering Graphics & Design, Information Technology, Business Studies

TECHNICAL PROFICIENCIES

Languages & Frameworks: Java • C++ • Python • C# • JavaScript • Dart • Node.js/Express • React.js • Flask • Flutter • HTML/CSS • SQL

Core Skills: Data Structures & Algorithms • OOP & Design Patterns • Full-Stack Development • RESTful API Design • Database Design & Optimisation • Software Architecture • Microservices • UI Test Automation • Agile/Scrum • Version Control (Git/GitHub/GitLab)

Cloud & DevOps: AWS • Azure • Google Cloud • Docker • Docker Compose • CI/CD Pipelines • Caddy/nginx • RabbitMQ

Tools & Platforms: VS Code • Visual Studio • Playwright • NUnit • Postman • PostgreSQL • Microsoft SQL Server • Linux/Unix • Advanced Microsoft Office

PROFESSIONAL EXPERIENCE

Technical Tester (Internship) | SITC *1 August – 24 August 2026*

UI test automation for enterprise cross-border shipping and customs management systems.

- Built UI test automation in C# using NUnit, Playwright for .NET and FluentAssertions, structured with the Page Object Model
- Converted manual test-case documents into automated tests for the Shipments module of an Angular-based shipping platform (grid, search, create, edit and validation flows)
- Automated the Shipments, Customs and Compliance dashboards of a second client system: 60+ passing tests covering status cards, charts and report navigation
- Created and maintained a test-case tracking sheet (97 test cases across 21 modules) to measure automation progress, and identified undocumented behaviour such as the OTP login step
- Kept test credentials out of the code using .NET User Secrets and managed the code with Git/GitHub

FNB Disruption 2025 Program Participant | First National Bank *2025*

Selected participant in FNB's innovation program focused on digital transformation and emerging technologies in the financial sector.

- Collaborated with industry professionals and technology innovators
- Analysed real-world technological challenges in modern banking systems
- Explored digital transformation strategies and fintech solutions
- Developed insights into cloud infrastructure and API integration in financial services
- Enhanced understanding of innovation, scalability, and modern software architecture

CERTIFICATIONS & ACHIEVEMENTS

AWS Training & Certification

- Kiro Getting Started – AWS Training & Certification (March 20, 2026)
- Spec-Driven Development with Kiro – AWS Training & Certification (March 21, 2026)

Workshops & Professional Development

- AI Agent Development Workshop – Hands-on practical training building and deploying an AI agent on a Linux (Ubuntu) server using OpenCode; covered agent configuration, tool integration, and deployment workflow (2026)

SOFT SKILLS & COMPETENCIES

Effective Communication • Critical Thinking & Problem-Solving • Teamwork & Collaboration • Leadership Abilities • Time Management & Organisation • Attention to Detail • Adaptability & Learning Agility • Work Ethic & Dedication

PERSONAL INFORMATION

Nationality: South African • Date of Birth: 26 June 2005 • Driver’s Licence: Yes • Own Transport: Yes • Languages: Afrikaans (Native), English (Fluent)